Target Assignment in Synthetic Tabular Data

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Abstract

1 Introduction

2 Related Work

3 Target Assignment in Synthetic Tables

We compare three constructions. A joint generator produces $(\widetilde{X}, \widetilde{Y})$; the generated-target baseline retains both. Joint relabeling retains exactly the same $\widetilde{X}$ and replaces $\widetilde{Y}$ with $h(\widetilde{X})$, where h is trained on real training data. Features-only relabeling fits a separate generator to X and assigns targets with the same labeler. We evaluate CTGAN and TVAE (Xu et al. 2019) and include the ongoing TabDDPM comparison (Kotelnikov et al. 2023).

3.1 Learning the Target Predictor

3.2 Synthetic-only and Mixed Training

4 Experimental Setup

4.1 Real-data Benchmark

The selected benchmark contains Adult, Census KDD, Covertype, MNIST12, MNIST28, News, and California Housing, with experiment seeds 42 and 43. Rows are partitioned before fitted preprocessing. Identical feature groups are kept together, split state 42 is shared across experiment seeds, and rare-class protection can change the realized partition sizes. Encodings, scaling, generators, and labelers are fitted on training data. Real development data select downstream checkpoints; real test data evaluate the selected checkpoint.

Generators use 500 epochs, batch size 500, and 100,000 synthetic rows. Mixed training concatenates all real training rows and all synthetic rows. RF, XGB, and DNN are target predictors. Neural networks evaluate downstream utility. The TabDDPM comparison uses unconditional joint and features-only models under this protocol; it is not a replication of the original paper's conditional classification protocol.

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4.2 Construction of the Simulated Datasets

The known-distribution benchmark separates prediction from recovery of the original joint distribution. Each dataset has independent training and test draws of 10,000 rows, and each synthetic table has 10,000 rows. Experiment seeds are 42 and 43. CTGAN and TVAE use 300 epochs and batch size 500. The seven datasets comprise three mixed continuous/categorical distributions and four categorical Bayesian networks.

Gaussian, Grid, and Ring Each row contains two continuous features X_0, X_1, a binary feature C, and a binary target Y. Gaussian draws X_0 and X_1 independently from $\mathcal{N}(0, 1)$. Grid independently selects each coordinate's center uniformly from $\{-4, -2, 0, 2, 4\}$, adds Gaussian noise of standard deviation 0.2, and divides by $\sqrt{8 + 0.2^2}$. It has 25 modes on a five-by-five grid. Ring selects one of eight equally spaced centers on the unit circle, adds isotropic Gaussian noise of standard deviation 0.2, and divides both coordinates by $\sqrt{0.5 + 0.2^2}$. Its coordinates are dependent.

All three use the same conditional feature and target rules:

$$C \mid X \sim \text{Bernoulli}(\sigma(2X_0X_1)), \tag{1}$$

where σ is the logistic function, and

$$h^*(X, C) = \mathbf{1}\{X_1 + [1 + 0.75(2C - 1)]X_0 > 0\}. \tag{2}$$

The observed target is $Y = h^*(X, C) \oplus E$, with independent $E \sim \text{Bernoulli}(0.10)$ and exclusive-or $\oplus$. Thus the Bayes accuracy is 0.90. Conditioning on C changes the decision boundary, as shown in Figure 1.

Categorical Bayesian Networks Asia, Alarm, Child, and Insurance use the packaged graphs and their original conditional probability tables. We sample ancestrally from each network, designate one node as the target (Table 1), and observe every other node as a feature. State codes are categorical, not ordinal. No additional label noise is introduced. Figure 2 shows Asia as an example. Oracle posteriors are computed by enumerating the target states with the other nodes observed. Bayes accuracy is estimated by averaging the maximum posterior over independent test features.

4.3 Evaluation Metrics

Real-data classification tables report the metric used in the dataset comparison: binary and macro F1 for Adult and

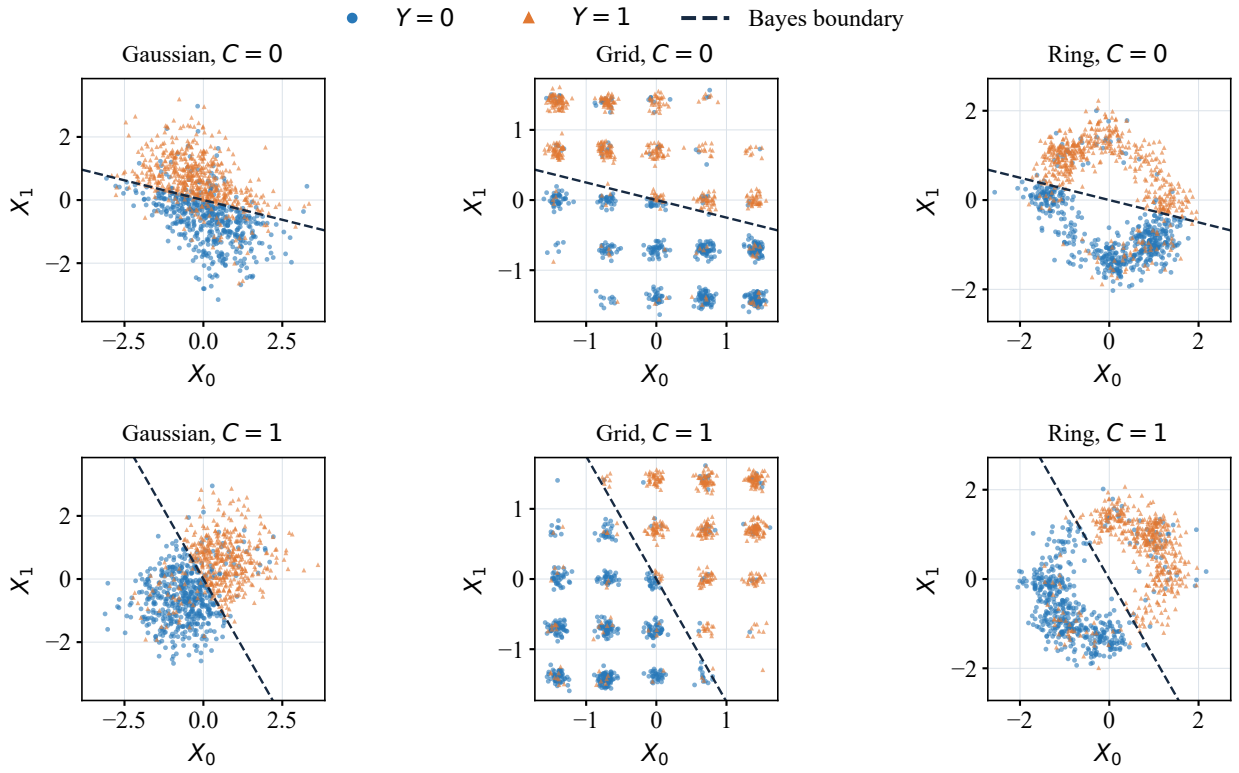


Figure 1: Gaussian, Grid, and Ring, conditioned on the binary feature C. Points are original-process rows; circles denote $Y = 0$ and triangles denote $Y = 1$. Dashed lines show the Bayes decision boundary. Each dataset is illustrated with 2,000 saved rows (seed 7), independent of the benchmark's 10,000-row draws with seeds 42 and 43.

Network	Features	Target	States
Asia	7	`dysp`	2
Alarm	36	`BP`	3
Child	19	`Disease`	6
Insurance	26	`Accident`	4

Table 1: Categorical Bayesian-network datasets. Every non-target node is observed. Original conditional probability tables define each joint distribution.

Census KDD, macro F1 for Covertype, and accuracy for MNIST12 and MNIST28. Regression tables report R^2 and D^2 absolute error. D^2 is one minus the test MAE divided by the MAE of the constant test-median predictor. Both metrics are unscaled; higher is better.

The simulated benchmark reports accuracy against observed targets, macro F1, and agreement with h^*. L_{syn} scores training rows under the known original joint distribution. L_{test} fits a normalized density to the training table and scores independent original test rows; it is a density-refit proxy. Bayesian-network likelihood reports use $\log(p + 10^{-8})$ once on the full joint probability and track exact-zero support violations separately.

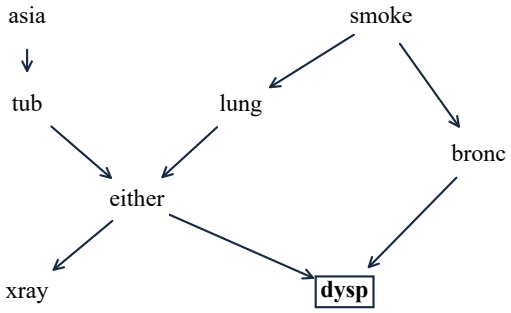


Figure 2: The Asia network. The boxed node, `dysp`, is the target; all other nodes are observed features. Arrows and conditional probability tables define the original joint distribution used to sample rows.

5 Results on Real Data

Tables 2 and 3 compare generated targets with X-only relabeling in the two training regimes. TabDDPM results are partial.

(a) Classification (%)

Method	Adult		Census KDD		Covertype	MNIST12	MNIST28
	Binary F1	Macro F1	Binary F1	Macro F1	Macro F1	Accuracy	Accuracy
Real only	67.45	**79.14**	**55.21**	<u>75.63</u>	50.47	**95.63**	**97.97**
Baselines: generated targets							
CTGAN	61.71	75.77	47.42	71.05	44.76	54.02	57.26
TVAE	56.74	72.93	47.00	71.21	43.42	92.83	94.20
TabDDPM	—	—	—	—	—	—	—
Our methods: X-only + predicted targets							
CTGAN-X + RF	65.63	77.87	52.75	74.66	54.16	88.83	87.95
CTGAN-X + XGB	66.43	78.06	54.24	75.45	50.48	90.28	88.83
CTGAN-X + DNN	**68.64**	<u>78.57</u>	<u>54.40</u>	**75.74**	**69.09**	91.14	91.95
TVAE-X + RF	65.86	77.98	48.53	72.21	46.21	92.84	94.55
TVAE-X + XGB	66.91	78.47	49.92	73.13	45.59	<u>94.20</u>	95.89
TVAE-X + DNN	<u>68.55</u>	78.39	52.85	74.79	<u>63.94</u>	93.80	<u>95.95</u>

(b) Regression

Method	News		Housing	
	R^2	D^2	R^2	D^2
Real only	-0.0290	-0.0445	0.5125	0.2822
Baselines: generated targets				
CTGAN	-0.0195	**0.0150**	0.6315	0.4136
TVAE	-0.0274	0.0049	0.7145	0.5054
TabDDPM	—	—	0.8020†	0.6045†
Our methods: X-only + predicted targets				
CTGAN-X + RF	-0.0003	-0.0006	0.7900	0.5857
CTGAN-X + XGB	0.0000	-0.0104	**0.8195**	<u>0.6212</u>
CTGAN-X + DNN	0.0018	<u>0.0074</u>	0.8039	0.6075
TVAE-X + RF	<u>0.0026</u>	-0.0026	0.7790	0.5732
TVAE-X + XGB	0.0020	-0.0319	<u>0.8173</u>	**0.6217**
TVAE-X + DNN	**0.0102**	-0.0208	0.7577	0.6061

Table 2: Synthetic-only training. Means over seeds 42/43; higher is better for every metric. D^2 uses absolute error relative to the test-median baseline. G-X fits generator G to features only, then uses the named labeler. Bold: best; underline: second-best distinct displayed mean, including real-only and ties. †: seed 42 only, excluded from ranking; —: no completed entry in the October 11, 2026, 00:25 Chicago TabDDPM snapshot. TabDDPM X-only results are unavailable. MNIST12 TVAE retains the feature-collision caveat.

5.1 Generated Targets versus Predicted Targets

5.2 Contrast Analysis

We evaluate two contrasts separately in each training regime. Relabeled minus generated compares predicted targets with generated targets, averaging equally over CTGAN/TVAE, RF/XGB/DNN, seeds 42/43, and both relabeled feature sources. Joint minus X-only compares the two feature sources with predicted targets in both arms, matching generator, labeler, and seed. TabDDPM is excluded from these contrasts because its matrix is incomplete.

Classification inference uses one macro-F1 contrast per dataset, with equal weight for all five tasks (Table 4). The two-sided paired t tests have four degrees of freedom. Confidence intervals are pointwise 95%, and p values are unadjusted. The contrast questions were selected after earlier analyses, so these tests are exploratory.

Regression tests use matched held-out predictions for each fixed task (Table 5). We average each case's loss over the specified fits before comparing constructions. Reductions in squared and absolute error correspond to gains in R^2 and D^2, respectively. We use 99,999 paired sign flips, giving identical-feature groups a common sign, and compare the absolute permuted and observed mean contrasts. The raw p value is $(b+1)/100{,}000$, where b counts permutations at least as extreme. These exploratory tests assume independent groups with sign-exchangeable loss contrasts under the null. They are conditional on the fitted models and do not estimate

(a) Classification (%)

Method	Adult		Census KDD		Covertype	MNIST12	MNIST28
	Binary F1	Macro F1	Binary F1	Macro F1	Macro F1	Accuracy	Accuracy
Real only	67.45	**79.14**	55.21	75.63	50.47	**95.63**	**97.97**
Baselines: generated targets							
CTGAN	65.17	77.69	44.69	68.42	49.61	71.88	92.77
TVAE	60.18	74.95	55.39	75.86	48.92	94.35	96.45
TabDDPM	—	—	—	—	—	—	—
Our methods: X-only + predicted targets							
CTGAN-X + RF	66.71	78.60	55.82	76.47	57.41	92.94	95.73
CTGAN-X + XGB	66.44	78.20	56.98	76.98	57.90	94.10	96.20
CTGAN-X + DNN	68.62	78.36	56.99	77.10	**74.76**	93.79	96.38
TVAE-X + RF	65.19	77.76	56.15	76.56	54.26	93.97	95.99
TVAE-X + XGB	66.84	78.56	**57.74**	**77.42**	55.60	94.80	96.87
TVAE-X + DNN	**68.97**	78.91	57.36	77.30	73.15	94.64	96.78

(b) Regression

Method	News		Housing	
	R^2	D^2	R^2	D^2
Real only	-0.0290	-0.0445	0.5125	0.2822
Baselines: generated targets				
CTGAN	-0.0173	0.0241	0.6356	0.4187
TVAE	-0.0219	**0.0360**	0.6997	0.5028
TabDDPM	—	—	0.7998†	0.5971†
Our methods: X-only + predicted targets				
CTGAN-X + RF	-0.0020	0.0080	0.7820	0.5877
CTGAN-X + XGB	**0.0023**	-0.0004	**0.8098**	**0.6157**
CTGAN-X + DNN	-0.0005	0.0203	0.8028	0.6083
TVAE-X + RF	0.0006	0.0019	0.7606	0.5551
TVAE-X + XGB	-0.0007	0.0086	0.8002	0.6061
TVAE-X + DNN	0.0014	0.0092	0.7875	0.6007

Table 3: Real + synthetic training. Metrics, means, notation, and highlighting follow Table 2. Mixing concatenates all real training rows and 100,000 synthetic rows. Daggers identify seed-42-only results.

variation from retraining.

6 Results under Known Distributions

6.1 Predictive Utility and Bayes Agreement

6.2 Joint-distribution Fidelity

7 Discussion

7.1 When Does Target Replacement Help?

7.2 Does the Generator Need to Model the Target?

8 Limitations

The two experiment seeds share real-data partitions, and MNIST12 and MNIST28 are related tasks. The corrected MNIST12 TVAE data retain the documented exact feature-collision caveat; train-only fitting does not establish that every test feature vector is unseen. Hard relabeling does not reproduce independent target noise. Density-refit scores depend on the specified estimators. The TabDDPM matrix is incomplete.

9 Conclusion

Training	Contrast	Gain (pp)	95% CI	Raw p
Synthetic-only	Relabeled − generated	+11.63	$[+1.33, +21.93]$	0.0350
	Joint − X-only	+0.36	$[-0.85, +1.56]$	0.4575
Mix	Relabeled − generated	+6.77	$[-0.31, +13.85]$	0.0566
	Joint − X-only	+0.14	$[-0.85, +1.13]$	0.7182

Table 4: Classification contrasts across Adult, Census KDD, Covertype, MNIST12, and MNIST28. Gains and pointwise 95% confidence intervals are in macro-F1 percentage points. Each dataset contributes one contrast, after averaging equally over CTGAN/TVAE, RF/XGB/DNN, and seeds 42/43. Raw p values use exploratory two-sided paired t tests across the five dataset contrasts ($df = 4$), with no multiplicity adjustment.

(a) R^2

Training	Contrast	News		Housing	
		Gain	Raw p	Gain	Raw p
Synthetic-only	Relabeled − generated	+0.0246	< 0.0001	+0.1208	< 0.0001
	Joint − X-only	-0.0031	< 0.0001	-0.0015	0.0336
Mix	Relabeled − generated	+0.0192	< 0.0001	+0.1212	< 0.0001
	Joint − X-only	-0.0010	0.0563	-0.0033	1.0000

(b) D^2 **absolute error**

Training	Contrast	News		Housing	
		Gain	Raw p	Gain	Raw p
Synthetic-only	Relabeled − generated	-0.0178	< 0.0001	+0.1448	< 0.0001
	Joint − X-only	+0.0038	< 0.0001	+0.0036	< 0.0001
Mix	Relabeled − generated	-0.0208	< 0.0001	+0.1382	< 0.0001
	Joint − X-only	+0.0025	< 0.0001	+0.0066	< 0.0001

Table 5: Regression contrasts with unadjusted, two-sided paired permutation p values (99,999 group sign flips). Gains are unscaled; positive values favor the first construction. CTGAN/TVAE, RF/XGB/DNN, and seeds 42/43 receive equal weight. Tests use squared error for R^2 and absolute error for D^2 on 7,929 News and 4,128 Housing test rows. Inference is exploratory and conditional on the fitted models; it does not measure retraining variability.

Task	Acc.	ΔAcc.	Δh^*	ΔL_{syn}	ΔL_{test}
Gaussian	82.06	+7.29	+9.09	+0.4081	-0.0195
Grid	86.27	+3.61	+4.39	+0.2823	-0.1906
Ring	87.49	+2.14	+2.70	+0.4261	-0.0581
Asia	85.48	+0.22	+0.41	+0.2354	-0.7728
Alarm	66.17	+14.72	+22.02	+0.4146	+0.0009
Child	82.42	+5.35	+7.92	+0.3816	+0.0563
Insurance	90.31	+4.24	+4.78	+0.1158	+0.0615

Table 6: Known-distribution benchmark. Acc. is generated-target accuracy (%), averaged over two generators and two seeds. Changes compare joint relabeling on the same features with generated targets: RF/XGB/DNN, two generators, two seeds (12 changes per task). Accuracy and h^* agreement changes are percentage points. Density changes use native scores; L_{test} is a refit proxy. BN scores use the full-joint epsilon convention.

References

Kotelnikov, A.; Baranchuk, D.; Rubachev, I.; and Babenko, A. 2023. TabDDPM: Modelling Tabular Data with Diffusion Models. In *Proceedings of the 40th International Conference on Machine Learning*, volume 202 of *Proceedings of Machine Learning Research*, 17564–17579. PMLR.

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